

A Wide Band Magnetotelluric Investigations to Delineate the Deep Crustal Geoelectric Structure in East Antarctica

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ABSTRACT

In the 25th Indian Scientific Expedition to Antarctica, wide-band magnetotelluric investigations were carried out during December 2005 to March 2006 in and around Schirmacher oasis. The aim of the study was to map the deep crustal geoelectric structure of east Antarctica around the Indian Research Station, Maitri. Limited long period MT studies were taken up earlier in Antarctica by other countries. Wide band MT data acquisition (0.001-1000 sec.) covering both AMT and MT signals became a problem related to get high quality data. The problems are tackled with computer aided online processing facility using internally heated laptop computer and special electrodes. A total number of six sites have been occupied along a profile oriented NE-SW direction to the south of Maitri with a station interval of 2-3 km with length of about 15 km. One site is situated on land mass in gneissic terrain close to Maitri and other five sites were to the south of Maitri on Continental ice sheet. MT data have been collected at each site for a duration of 4-5 days to acquire long period signals and also to obtain good quality short period signals. Use of titanium electrodes as e-probes has reduced the contact resistance to kilo-ohms and facilitated recording of both high and low frequency signals. In the present study, the logistics of field data acquisition procedures of the wide band magnetotelluric study are presented. A deep crustal geoelectric section obtained from 2-D modeling results showed a thin (10 km) high resistive upper crust followed by less resistive mid-lower crust. A feeble resistivity contrast is also observed as a boundary between lower crust and upper mantle, perhaps represents an electrical Moho. Results are compared with earlier (24th Indian Antarctic Expedition) MT results and discussed.

INTRODUCTION

Antarctica has been broadly divided into eastern and western parts by transantarctic mountains (Gregory, 1901). West Antarctica encompasses

the Scotia arc region, the Antarctic Peninsula, Ellsworth Land, the Ellsworth mountains and Marie Byrd Land, while East Antarctica consists of Dronning Maud Land, Enderby and Kemp Lands. An approximately 95 per cent of its surface is covered with thick (1–3 km) ice, and so crust and mantle structure must be deduced geophysically. Geophysical deductions must make use of only a handful of physical properties of the Earth, one of which is its electrical resistivity. The geophysical method that can provide images of resistivity to deep crustal or upper mantle depths is magnetotellurics (MT) (Vozoff 1991; Harinarayana et al., 2003, Wannamaker et al., 2004). This physical property in turn can provide information on primary structures (sedimentary distributions, lithologic contrasts, major fault offsets), geochemical fluxes (hydrothermal alteration, remobilized graphite and sulphides) and thermal regime (prograde or melt-exsolved fluids, crustal or upper mantle melts, mineral semi-conduction).

The Queen Maud Land area in East Antarctica, in Indian Research Station 'Maitri' is situated in Schirmacher Oasis. Schirmacher Oasis is a small ice-free land mass of 35 km area, located at the intersection of coordinates 70.8° S lat. and 11.8° E long. East Antarctica has been accepted as a stable cratonic block of Pre-Cambrian rocks, but more recently is thought to have been a component of a Late Proterozoic supercontinent Rodinia (e.g. Dalziel 1991; Borg and DePaolo 1994; Rodgers et al. 1995; Satish-Kumar et al., 2008). West Antarctica has been assembled to East Antarctica from accreted blocks and subduction-related plutonism since lower Paleozoic time (Storey & Alabaster, 1991).

Earlier Studies in the Region

Geophysical and geological surveys along and across the continental margin of Antarctica play a major role in deciphering the Gondwana puzzle. Geophysical survey measurements are of particular importance in Antarctic research because of the dominating ice cover of this continent. Global Geoscience Transect projects were carried out from different countries in this part of Antarctica. Previously, Geomagnetic, deep seismic soundings, gravity and magnetic studies were carried out both on exposed land mass as well as on the ice covered regions (Dimitriev & Kochi, 1964; Kogan & Stroev, 1972).

The seismic refraction studies (Bentley, 1991; Behrendt, 1999; Bentley & Clough, 1970) and Radar soundings (Jiracek, 1967; Bentley, 1991) were

restricted to west Antarctica and part of east Antarctica. Magnetotelluric studies were carried out (Hessler & Jacobs, 1966; Beblo & Liebig, 1990; Fournier, 1994; Kong et al, 1993, 1994; Wannamaker et al, 2004) only in part of West Antarctica and in Trans Antarctica mountains at relatively long periods.

Wide-band MT studies were carried out on the Continental ice sheet to the south of Schirmacher Oasis along NE- SW profile during December 2005 to March 2006 in the 25th Indian Antarctic expedition. In this study, we are presenting the results of deep crustal geoelectric structure in east Antarctica. Earlier a geo-electrical survey was carried out aimed at obtaining information about thickness of the weathering layer in the Schirmacher Oasis (Bhattacharya et al. 1987).

Crustal thickness of West Antarctica varies between 20 and 30 km in contrast to ~40 km in East Antarctica, derived on the basis of seismic refraction profiles near its NE coast, and gravity interpretations region-wide (Bentley, 1991). However, the Moho depth in Schirmacher Oasis region has been interpreted to ~35 km on the basis of geomagnetic data (Wagner and Lindner, 1991). Surface wave studies confirm separateness of East and West Antarctica, with the former showing more craton-like velocities on an average and with the highest lateral gradient in model velocities (Bentley, 1991; Danesi and Morelli, 2000; Ritzwoller et al., 2001; Harley, 2003). Seismicity throughout Antarctica is low, which is consistent with a slow moving plate almost entirely surrounded by mid-ocean ridges. However, the western part is approximately 10 times more active than the east (Bentley, 1991, Hole and Lemasurier, 1994). Electromagnetic measurements have been conducted for bed rock investigation (Bhattacharya and Majumdar, 1997; Bhattacharya et al., 1987). The influence of mantle plume in Antarctica has been reported by many authors (Wannamaker et al., 2004). The upwelling mantle plume, in general, tries to pass through mobile belts, which are relatively thin as compared to relatively thick cratonic parts (Raval and Veeraswamy, 2000). However, particularly the study of plume-related magmatism by Sushchevskaya et al. (2008) within Antarctica proves the spreading of the Karoo-Maud plume, which determined the split of the Gondwana continent and formation of the Indian ocean, to the east – from Queen Maud land towards Schirmacher Oasis. Isotopic compositions of dolerite dykes of Schirmacher Oasis clearly indicate the crustal contamination processes which took place during the plume upwelling and emplacement into the continental crust. The earth's

most extensive origin, i.e. Mozambique belt that was formed during the collision of east and west Gondwana seems to extend from South Africa in the north to Dronning Maud land in the south (Jacobs, 1999; Ravikant et al., 2004, 2007). According to Satish-Kumar et al. (2008) the Schirmacher Oasis could be part of SE Africa, where as the inland mountains were part of crust formed during the final amalgamation of East Gondwana.

Geology of Schirmacher Oasis

“Schirmacher Oasis” is located near the Princess Astrid Coast, Central Queen Maud Land, East Antarctica (Fig. 1). This area ($70^{\circ}44'-70^{\circ}47'S$ and

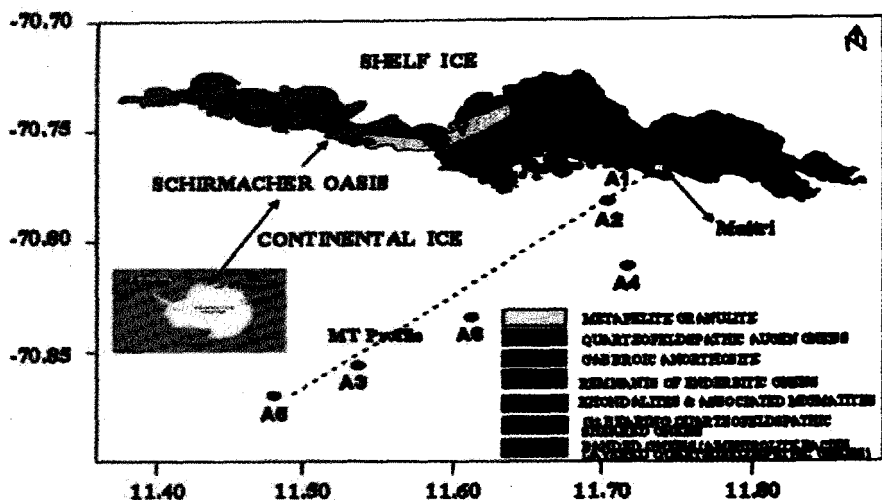


Fig. 1: Geological map with MT site locations in Schirmacher Oasis, East Antarctica

$11^{\circ}25'-11^{\circ}55'E$) of about 35 km^2 is situated parallel to the E-W trending coast line between the inland ice and the shelf ice (Tingey, 1991). The Precambrian crystalline basement of Schirmacher Oasis is part of the East Antarctic shield. The rocks of Schirmacher Oasis were affected by an initial granulite facies metamorphism in Proterozoic times (Sengupta and Bose, 1997; Hoch and Tobschall, 1998). The main deformation was followed by an amphibolite facies metamorphism which was accompanied by migmatization and isoclinal folding of the rocks (Kampf and Stackebrandt, 1985). The formation of east-west striking overthrusts took place subsequently. The Pan-African thermo tectonic activation at the time of Precambrian-Phanerozoic (600–450 Ma) overprinted the crust and caused faulting, dynamo metamorphism under amphibolite facies P-T conditions and dyke intrusions (Ravich and Krylov, 1964). The youngest fault event

is connected with the formation of cataclasites and greenschist facies metamorphism in the area of faults (Kampf and Stackebrandt, 1985). The metamorphic complex of Schirmacher Oasis is intersected by numerous lamprophyre, basalt, pegmatite and aplite dykes. On the basis of field observations it has been indicated that the minettes of Schirmacher Oasis are younger than the pegmatites and older than the basalt dykes. Kaiser and Wand (1985) and Wand et al. (1988) distinguished two age groups of basalts, Palaeozoic and Mesozoic. Conventional K-Ar isotope age determinations yielded isochrons of 223 and 354 Ma. Potassium feldspars from pegmatites gave Pb-Pb ages between 600 and 865 Ma (Bielicki et al., 1991). The studied minettes are obviously older than 354 Ma, but younger than the pegmatites. Whole rock-biotite Rb-Sr isotope isochrones indicate ages of 430 Ma and 703 Ma for the minettes (Hoch and Tobschall, 1998).

Schirmacher Oasis forms the foot hill zone of the mountain ranges of the Queen Maud Land and lies north of the Wohlthat Mountain ranges. The Schirmacher oasis belongs to the east Antarctica platform of precambrian and magmatic rocks (Kaul et al., 1985; Sengupta, 1986 and Singh, 1986). These rocks have undergone several stages of metamorphism, magmatism and deformation. The main lithological units are charnockite, granitic gneisses, gneissose migmatites of different varieties, relicts of metabasic rocks, calc-silicates and aluminous gneisses. These lithological units have an ENE trend intersecting the hill range at low angle. The major portion of the eastern Schirmacher Oasis displays prominent banding with considerable compositional difference. The darker varieties grade into pyroxene granulite and are essentially composed of biotite with some amphibolites. Alaskites are light grey or off-white in colour without distinct colour banding and contain variable amounts of garnet. The garnet biotite gneisses are characterized by the presence of large elongate clots of garnet and biotite. The calc-gneiss, khondalites and associated migmatites unit is layered with individual layers varying from a few centimeters to a few meters. The colour banding in the calc-gneiss is partly represented by the variable calc-silicate assemblages in different layers and partly by lit-par-lit emplacement of the granitic material. Banding is accentuated by thin zones of mylonite which alternate with the less sheared rocks.

The Schirmacher Oasis Metamorphic Complex consists of metamorphics, migmatites and alternations of the rock assemblages mentioned. Additionally, there exist dyke rocks such as metabasalt, dyke basalt, lamprophyre, microgabbro, pegmatite and aplite as well as tectonites

such as blastomylonite, cataclasite, mylonite, granulite and S-tectonite (Kamp & Stackbrandt, 1985).

Magneto Telluric Method

In Magneto telluric (MT) method, naturally occurring electromagnetic (EM) wave fields have been used as sources for imaging the electrical resistivity structure of the Earth (Vozoff, 1991). The incident EM waves propagate vertically downward and usually are treated as planar in geometry.

In the conducting Earth, EM waves at typical frequencies of the method (<1000 Hz, for example) travel diffusively, such that high frequency (short period) waves penetrate a relatively short distance while low-frequency (long period) waves can reach the mantle depths. The electric (**E**) and magnetic (**H**) vector fields scattered by buried structure and measured at the surface may have arbitrary polarization relative to the incident fields. This requires a tensor relationship between the measured fields as a function of frequency (or period T , its inverse) denoted for the horizontal components through

$$[\mathbf{E}] = [\mathbf{Z}][\mathbf{H}],$$

where $\mathbf{Z}$ is the 2×2 complex impedance. The individual impedance elements typically are transformed into an apparent resistivity (ρ_a) and an impedance phase (ϕ) for presentation and modeling. On a uniform half-space of resistivity ρ , $\rho_a = \rho$ and $\phi = 45^\circ$. For more complex structures, ρ_a versus T approximates a smoothed version of ρ versus depth below the measurement. The phase ϕ tends to be proportional to slope of ρ_a versus T , and thus is more reflective of spatial gradients in the subsurface resistivity. These primary data in turn are transformed to constrained resistivity models through the process of inversion, specified in the model construction section below.

Over 2-D structures where one of the measurement axes is parallel to geoelectric strike (x here by convention), the diagonal entries of $\mathbf{Z}$ are zero (Vozoff, 1991). In this situation, the MT response separates into two independent modes. These are the transverse electric (TE) mode, where $E_x = Z_{xy}H_y$ and electric current flows parallel to strike (x -axis), and the transverse magnetic (TM) mode, where $E_y = Z_{yx}H_x$ and current flows perpendicular to strike (y -axis). Additionally, a 1×1 tensor $\mathbf{K}_z$ relates the vertical and horizontal magnetic fields through

$$[H_z] = [\mathbf{K}_z][\mathbf{H}].$$

For a 2-D Earth, $K_{zx} = 0$ and K_{zy} reflects cross-strike changes in current flowing along strike (TE mode).

Earlier MT Studies in Antarctica

Hessler and Jacobs (1966) recorded analogue, long period ($T > 1$ min) E-fields over glacial ice at Vostok with 200 m bipoles and copper screen electrodes buried in brine-soaked firn. Correlations were visible with simultaneous magnetic records, but large spike noise was common and no MT impedances were estimated. MT measurements were taken by Fournier (1994) on the Antarctic peninsula but electrical contact either was on exposed earth or sea ice of similar resistivity (1000 ohm m) and so not characteristic of the continental interior. This also appears to have been the case in the surveys of Kong et al. (1993, 1994) on the peninsula of West Antarctica and near the Larsmann Hills of East Antarctica. Good-quality MT data at relatively long periods (20–2000 s) has been acquired by Beblo and Liebig (1990) over thin glacial ice cover in North Victoria Land. They used copper screens as electrodes over a 25 m bipole span and an electrometer of very high input impedance. Signals in this longer period range are due to high amplitude ionospheric micro-pulsations, but the lack of shorter period data makes it difficult to quantify resistivity of most of the crustal column. However, Wannamaker et al. (2004) did show that sounding results during both low- and high-activity times of the polar electrojet were very similar, from which they concluded that non-plane-wave source effects were not a serious issue.

The present Magneto telluric studies were carried out on the land mass of Schirmacher Oasis for the first time in part of East Antarctica by MTS group, NGRI, Hyderabad in the 24th Indian Antarctic Expedition during December 2004 to March 2005.

MT Investigations at Schirmacher Oasis, East Antarctica

During present study, MT data was collected using the GMS-05 and ADU-06 systems of Metronix, Germany with a station spacing of ~2-3 km. The average recording time per site was approximately 4-5 days. This is slow, compared with most MT surveys on land because of the additional logistic requirements of Antarctica. MT data have been collected in the period range of 0.001–1000 sec along NE-SW oriented profile to the south of Schirmacher Oasis, East Antarctica (Fig.1). The length of the profile is about 15 km covering 6 sites. Out of six sites, the first site (A1) has been

occupied over a gneissic terrain (on exposed land mass) at about 2 km west of Maitri, and the remaining five sites were occupied to the south of Maitri on the Continental ice sheet. Here, the conventional electric field sensors (porous pots) are not suitable for measurements. So, Titanium sheets ($60 \text{ cm} \times 60 \text{ cm} \times 3 \text{ mm}$) have been used as electric field sensors, and it has been buried in the ice in a vertical direction. Bentonite powder has been used for better contact between the titanium sheet and the ice. High strength electric cables (multi-stranded) were used to avoid breakup/cracks of wire at low temperatures ($< -10^\circ\text{C}$). The field activities in MT measurements in Antarctica are shown in Figs 10a-10d.

Despite various difficulties such as extreme low temperature, snow storms (blizzards) etc., patient surveying led to the acquisition of good quality MT data. The apparent resistivity and impedance phase response for all the soundings, after Robust processing (Egbert, G.D. & Booker, J.R., 1986), (Huber, P.J., 1964) for site no. A1 and A3 is shown in Fig. 2. The most striking feature is the variation of apparent resistivity in both TE and TM modes, i.e. high resistive layer (at short periods) followed by conductive layer at long periods. The pseudo-sections of apparent resistivity

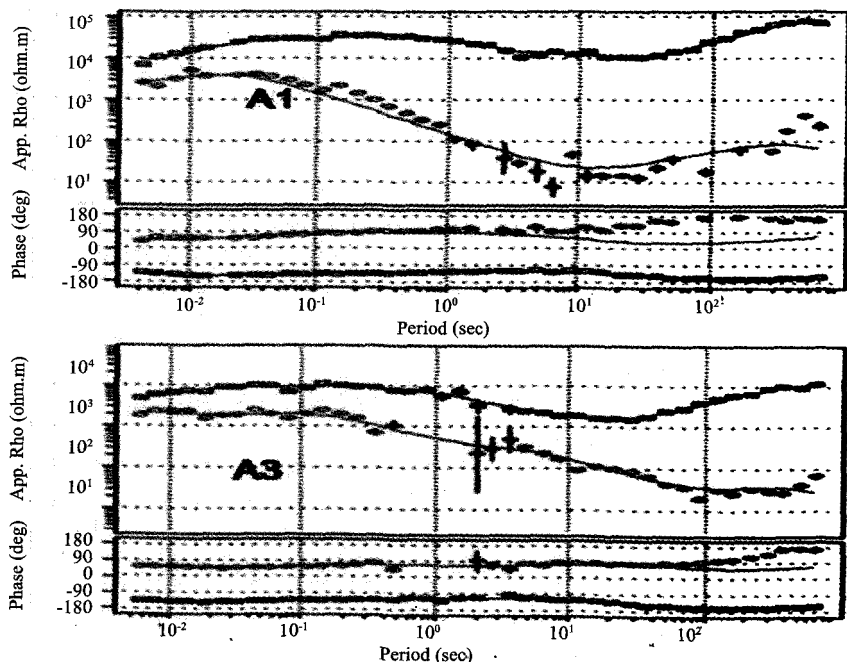


Fig. 2: A typical MT curve of apparent resistivity and phase response for site no. A1 and A3

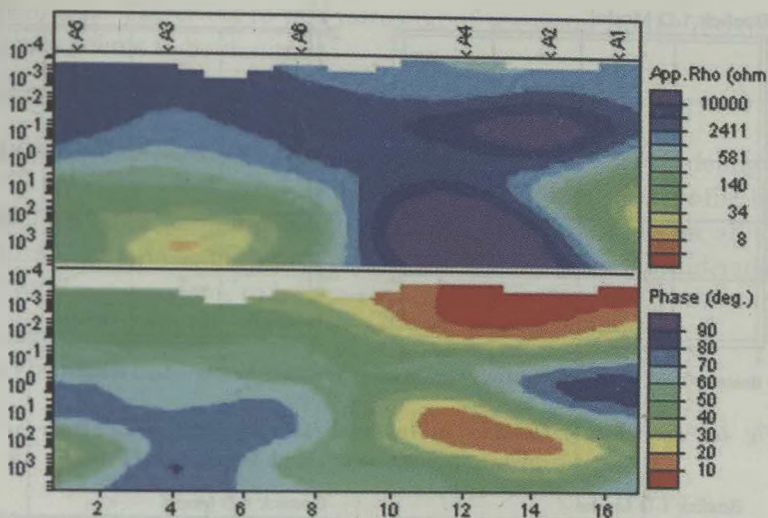


Fig. 3a: Pseudo section of Apparent Resistivity & Phase (TE-mode)

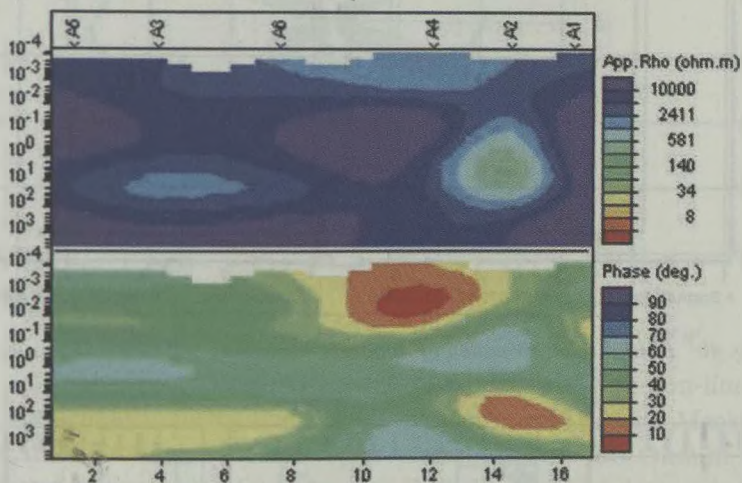


Fig. 3b: Pseudo section of Apparent Resistivity & Phase (TM-mode)

and phase for TE-mode and TM-mode are shown in Fig. 3a and 3b. The Bostick transformation of apparent resistivity has shown true resistivity as a function of depth in Fig. 4a and 4b. The induction arrows for 3 frequencies in Fig. 5 have shown the average direction as N23°E.

Schirmacher Oasis is an exposure surrounded by shelf ice towards northern side possibly with sea water below the ice and continental ice on

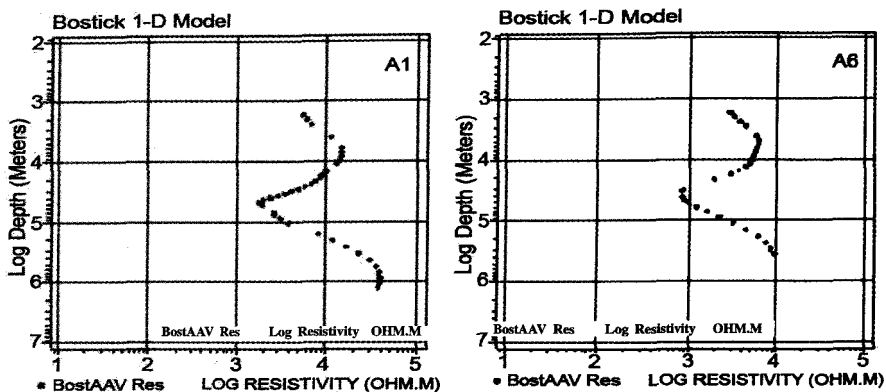


Fig. 4a: Bostick transformation of Apparent Resistivity and Phase shows the true resistivity as a function of depth for site no. A1 & A6

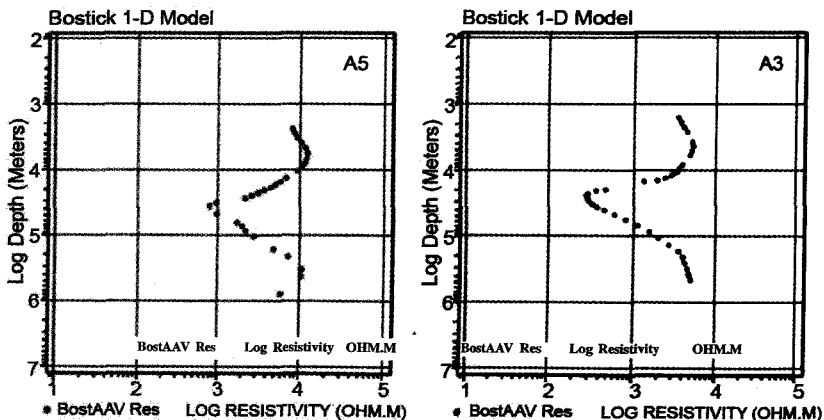


Fig. 4b: Bostick transformation of Apparent Resistivity and Phase shows the true resistivity as a function of depth for site no. A5 and A3

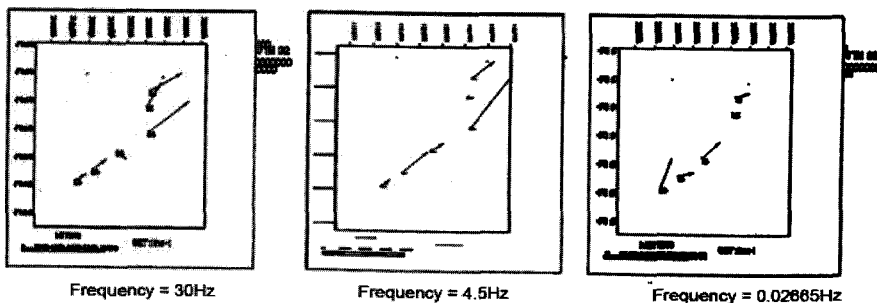


Fig. 5: Induction arrows for different frequencies shows N230E direction

the southern side. The effect of this intrusion of sea water on the response of MT curves seems to be minimum.

2-D Modeling

MT data for five sites have been considered here for 2-D modeling as site no. A2 is noisy. Geological strike for the sites along the profile has been computed using Groom-Bailey decomposition technique for all the sites. A rose diagram has been prepared after 90° ambiguity consideration and $N67^\circ W$ direction seems to be the regional strike (Fig. 6).

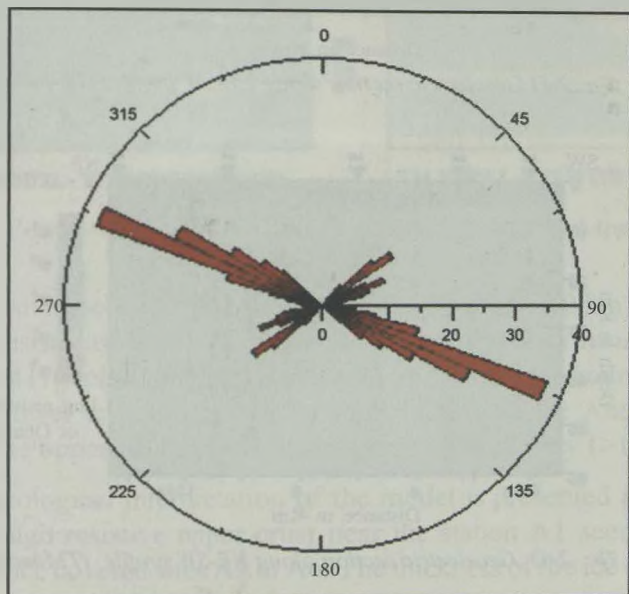


Fig. 6: Groom-Bailey decomposition shows the regional strike as $N67^\circ W$

MT data is subjected to 2-D inversion modeling using non-linear conjugate gradient as implemented WinGLink platform (Rodi and Mackie, 2001). An initial model with half space resistivity of 100 Ohm.m. is considered and subjected to 150 iterations. Both TE and TM data with six decades (0.001 to 1000 sec.) are considered here with an error floor of 5% for resistivity and phase. The value for tau of 3 is used for smoothness of the model. In order to test the model for consistency, different options are considered and the inversion is repeated. Thus the model derived considering TE only, is presented in Fig. 7a and TM only, is presented in Fig. 7b. Later considering both TE+TM, the derived model is presented in Fig. 7c. RMS error 2.03% is obtained for TE+TM mode for the final model presented in Fig. 7c.

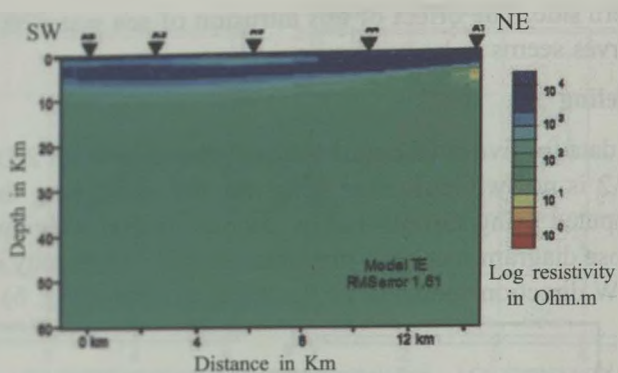


Fig. 7a: 2-D Geoelectric section along NE-SW profile, (TE-mode)

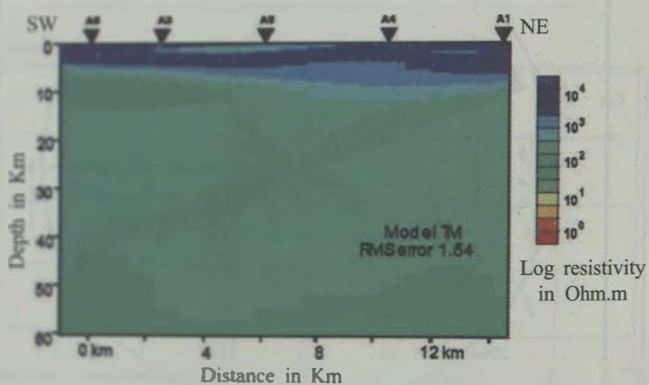


Fig. 7b: 2-D Geoelectric section along NE-SW profile, (TM-mode)

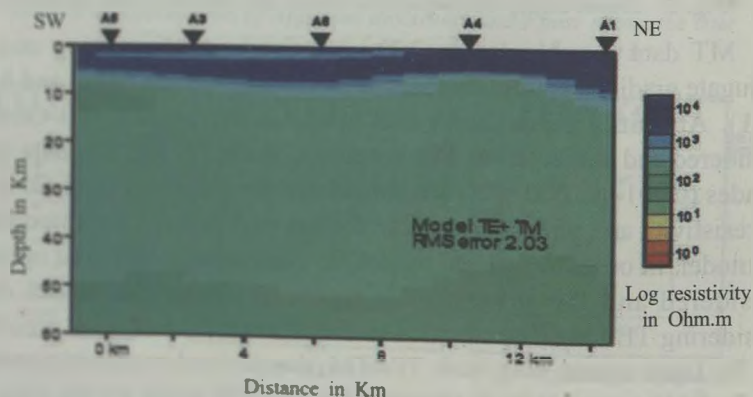


Fig. 7c: 2-D Geoelectric section along NE-SW profile, (TE+TM-mode)

An example of fit between the observed and computed data for this profile in the form of apparent resistivity and phase pseudo sections are presented in Fig. 8 for both the TE and TM-mode. A reasonably good fit is observed for all stations.

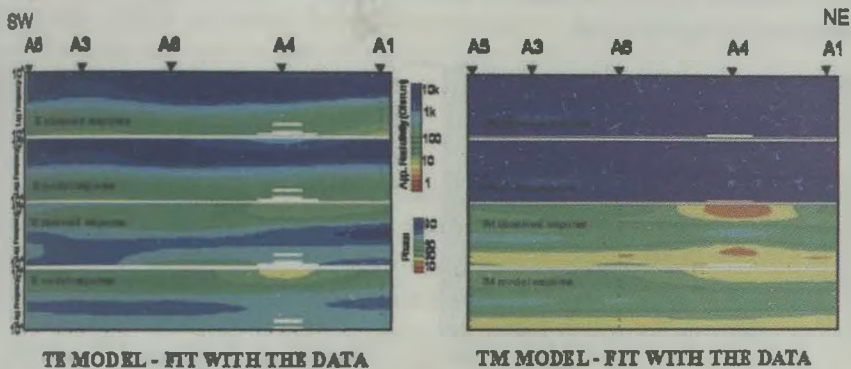


Fig. 8 : Pseudo-sections of 2-D model for observed and modeled App. Resistivity & Phase fit of TE & TM-mode

As can be seen from the model, the upper crust (<10 km) is dominated by high resistivity layer (10^4 ohm.m) followed by less resistive mid and lower crust (10^2 ohm.m). Unlike other shield regions, the boundary between mid and lower crust is not clearly visible below the Antarctic region. Further, the upper mantle tend to show higher resistivity ($>10^2$ ohm.m).

A geological interpretation of the model is presented in Fig. 9. The exposed high resistive upper crust near the station A1 seem to continue below the ice covered sites A3 to A6. The thickness of the ice cover exhibits

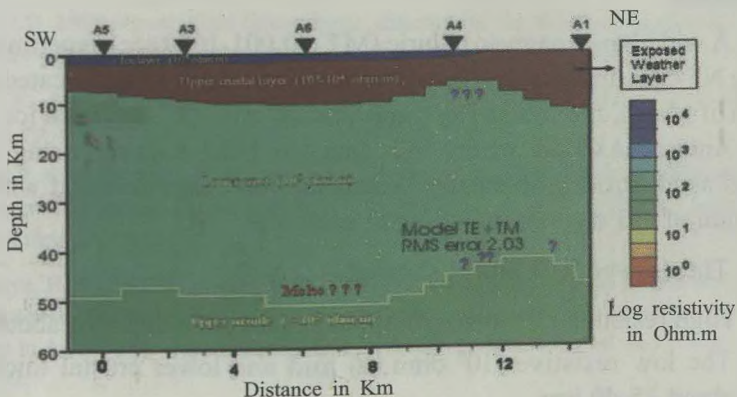


Fig. 9 : Geological interpretation of the geoelectric model

in the form of high resistive layer seems to be varying from about 50 m near the site A4 and gradually increasing towards SW and attains a maximum thickness of nearly 800 m. The mid, lower ($>10^2$ ohm.m) and upper mantle ($\sim 10^2$ ohm.m) is seen clearly with distinct resistivity as compared to upper crust.



Fig. 10a: MT site layout and preparation at site no. A1



Fig. 10b: Unloading of MT equipment at site no. A3



Fig. 10c: MT Equipment – GMS 05 at site no. A3



Fig. 10d: MT data acquisition at site no. A3

Summary

A wide-band magneto telluric (MT) (0.001-1000 sec) experiment made along NE-SW oriented profile with a length of about 15 km located towards south of Maitri, the Indian Research Station on the Continental ice sheet of East Antarctica. Good quality MT data has been acquired using titanium sheets as electrodes, internally heated lap-top computers and with a long duration of 4-5 days of recording at each site.

The derived 2-D model showed –

- High resistive (10^4 ohm.m) upper crust upto a depth of about 10 km.
- The low resistive (10^2 ohm.m) mid and lower crustal thickness of about 35-40 km.
- Moderately resistive ($>10^2$ ohm.m) upper mantle below 40 km.

Resistivity boundary is observed at a depth of 40-50 km, perhaps indicative of an electric Moho. However detailed and independent information from other geophysical data is necessary to confirm our results. The model determined from the present study is different from the 2-D resistivity model of South Pole (Wannamaker et.al, 2004). This indicates that the structure near South Pole and near Schirmacher Oasis may have different characters.

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